Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829 · LINE solaress1
www.solaress.org · www.โซลาร์เอสส์.com

Case Study: PV Maintenance at a Nursing Home

A nursing home in Lampang · 25 kWp rooftop, 20 kW 3-phase inverter · Plant code SG-52000-20-01

Site visit: 14 November 2026 · PV cleaning, inspection, thermal scan and current checks



Aerial view of the rooftop PV system

System OK
Solar generation working normally

3 issues
Found that need the customer's action

≈ 1,450 THB
Lost every month to the leakage

Summary of findings

#	Finding	Priority	Recommended action
1	**Cracked PV module** – front glass shattered by an outside impact (not covered by warranty)	MEDIUM	Replace with a matching module, or remove it from the string
2	**Hot neutral** – 1-phase aluminium 25 sq.mm cable at the old meter, up to 65.4 °C	HIGH	Re-terminate or replace the cable; monitor loads meanwhile
3	**Leakage current 1.91 A** on the 1-phase supply – wasted energy and shock risk	HIGH	Trace and repair the leaking circuit; confirm RCD protection

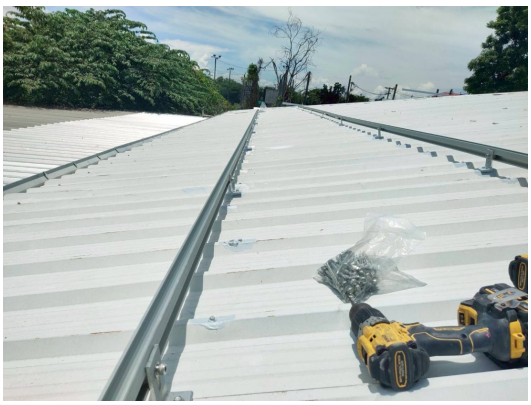
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The system as installed (September 2025)

Solar ESS installed this system in September 2025. These installation photos are the reference for today's checks: the same mounting, inverter, protection and phase currents can be compared visit to visit.



Rooftop mounting work



Rails fixed to the metal roof



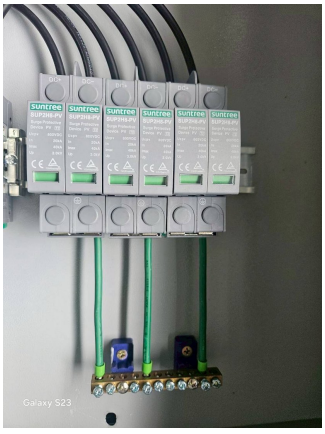
Inverter with AC and DC boxes



Inverter and protection boxes, doors open



AC box: main breaker and outgoing breaker



Surge protection devices (SPD) and earthing

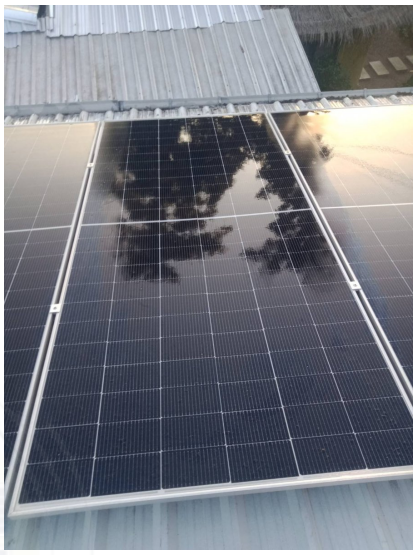
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Work done today

The team cleaned every PV module, inspected each one for damage, scanned the cables and meter connections with a thermal camera and measured currents with a clamp meter. Dust and bird droppings were removed, which restores light reaching the cells.



Before: dust and dirt build-up on the modules



After: modules cleaned



Cleaning with water and a soft long-handled brush



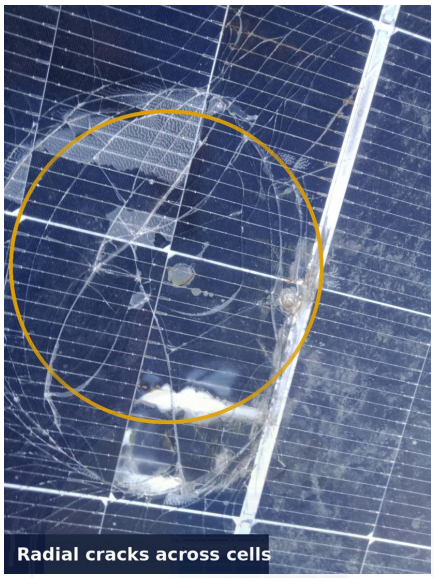
Array after cleaning

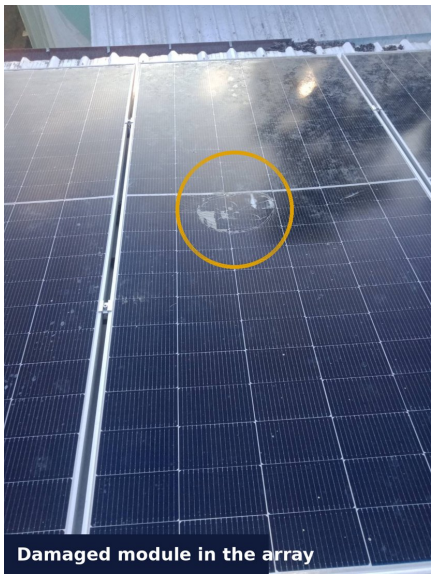
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Finding 1 · Cracked PV module

One module has shattered front glass. The cracks spread out from a single impact point, which shows the damage came from an outside object (thrown or falling), not from an internal cell crack or a circuit fault. **The manufacturer's warranty therefore does not cover it.**







Left: impact point. Middle: cracks running across the cells. Right: the damaged module in the array.

Why it matters

Broken glass lets moisture into the cells. Over time this lowers output, can create a hot spot and can cause an insulation fault that trips the inverter, which stops the whole string. The module should not stay in service long-term.

Options for the customer

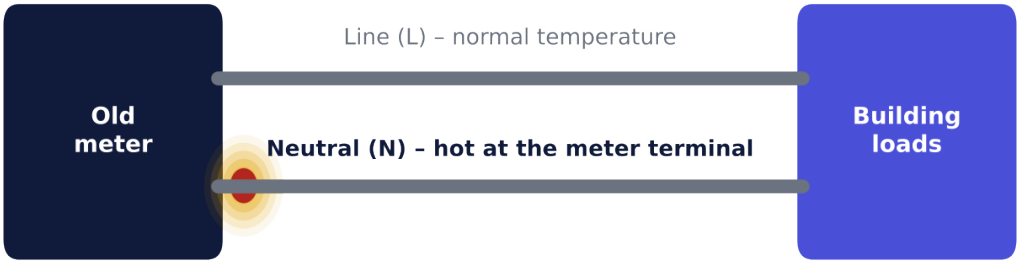
Option	What we do	Cost	Effect on the plant
A. Replace (recommended)	Fit a new module of the same model and rating so the string stays matched	1 module + installation (quoted separately)	Full output restored; risk removed
B. Remove	Take the module out and re-wire the string to bypass it	Labour only	Plant loses that module's output; string voltage checked against the inverter range

Until it is fixed
Solar ESS will watch this string on SolisCloud for insulation or string-current alarms.

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Finding 2 · Hot neutral cable at the old meter

The thermal scan showed the **neutral** of the single-phase aluminium 25 sq.mm cable to the old meter running hotter than the line conductor. Today it reached up to 65.4 °C. The system still runs, but the cause should be fixed soon before it gets worse.

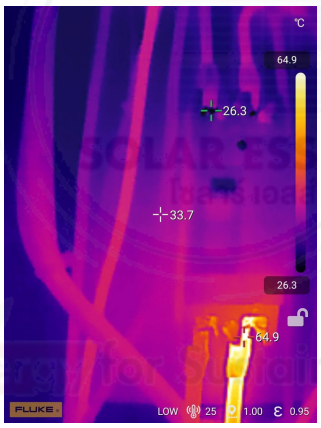


1-phase aluminium 25 sq.mm: loose / oxidised joint → resistance → heat

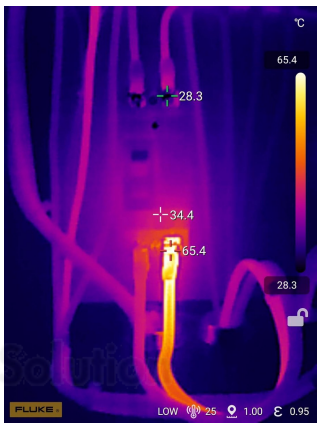
Today's thermal scan (14 November 2026)



38.9 °C at the terminal (breaker body 35.4 °C)



64.9 °C at the neutral terminal



65.4 °C at the neutral terminal and cable

Close to the cable limit
The neutral terminal reached 65.4 °C – about 37 °C above the surrounding parts (28.3 °C) and close to the 70 °C rating of standard PVC-insulated cable. The reading changes with the load, so at peak load it can run hotter. We recommend re-terminating it soon rather than only monitoring.



Meter area at the site (installation photo, Sept 2025). The hot neutral was found on the 1-phase aluminium 25 sq.mm cable feeding the old meter. Re-terminating it is quick work during a planned short shutdown.

Why a neutral runs hot

In a single-phase circuit the line and neutral carry the same current, so they should run at about the same temperature. A hotter neutral usually points to a **loose, corroded or oxidised termination** – common with aluminium – or to the neutral carrying return current from other circuits. A poor joint heats up more under load, which loosens it further over time.

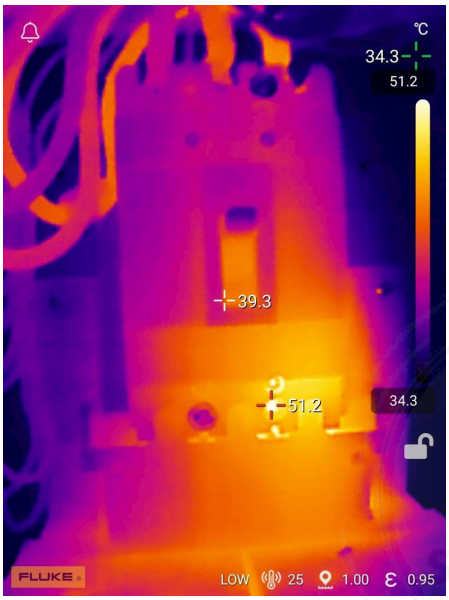
Options

Option	What we do	Result
A. Re-terminate or replace (recommended)	New bimetallic (Al–Cu) lugs with anti-oxidant compound, torqued correctly; or a new cable sized for the load	Removes the cause
B. Monitor the loads	Spread heavy loads (air conditioners, water heaters, pumps) over the day and re-scan at the next visit	Manages the risk only

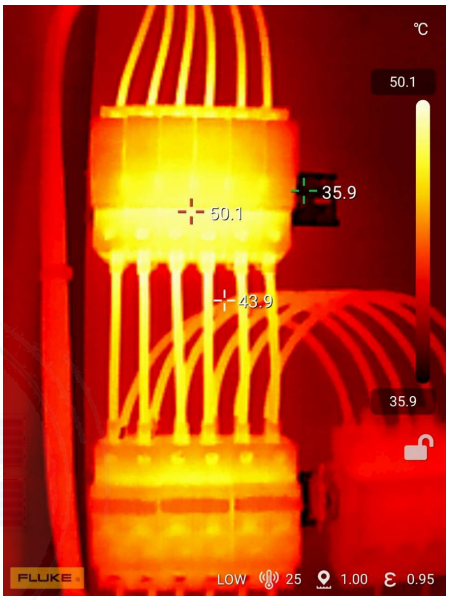
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Thermal scan reference – at installation (September 2025)

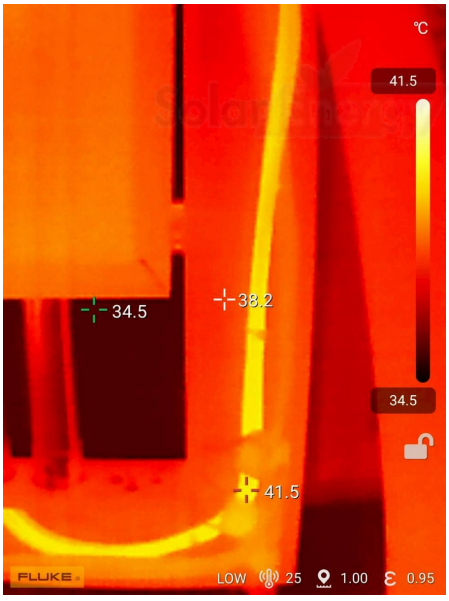
A thermal camera shows where current meets resistance: loose, oxidised or overloaded connections run hotter than the parts around them. These scans were taken when the system was installed in September 2025 and serve as the baseline for later scans. A leak of about 2 A to earth gives too little heat to see on its own, so the leak itself is located with the clamp meter and an insulation test; the thermal scan checks that the connections carrying the load are not overheating.



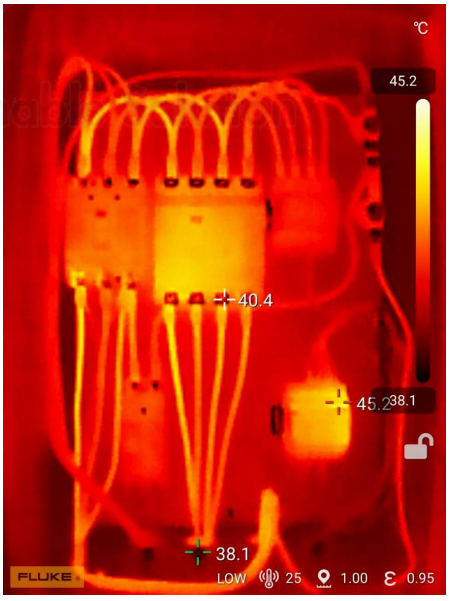
Breaker terminal: warmest point 51.2 °C (body 39.3 °C, background 34.3 °C)



Terminal block / SPD connections: 50.1 °C (cables 43.9 °C)



Cable run at the box entry: hot spot 41.5 °C at the bend (box 34.5 °C)



Distribution box overview: up to 45.2 °C

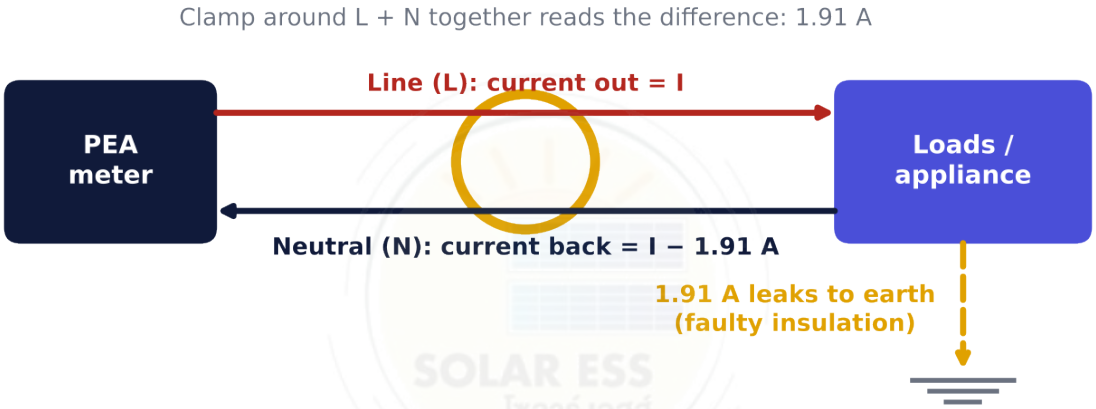
What we will do
Compare these baseline readings with today's scan, re-torque the warmest terminals together with the neutral repair, then re-scan the same points after the leakage repair and compare the temperatures.

Finding 3 · Leakage current on the single-phase cable



The clamp meter, placed around the line and neutral together, read **1.91 A**.

In a healthy circuit the current going out on the line comes back on the neutral, so this reading should be close to zero. About 2 A is escaping through another path, most likely to earth through faulty insulation.

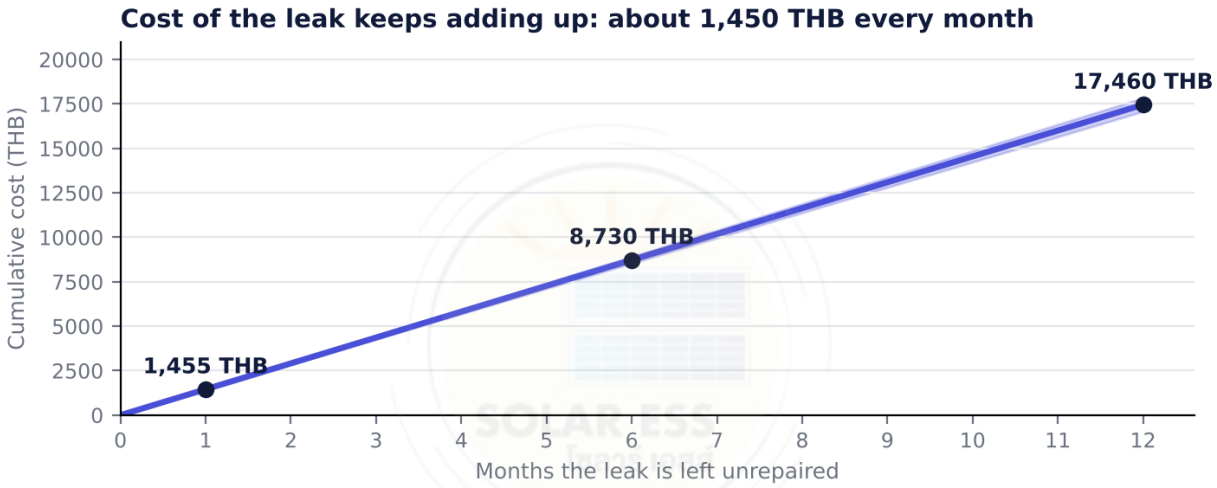


How the clamp reading reveals the leak

What the leakage costs

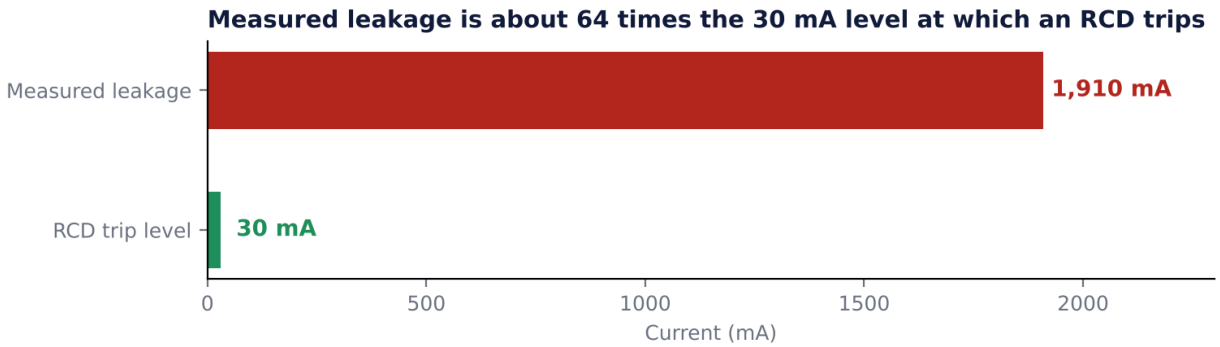
Leakage power = 230 V × 1.91 A ≈ **439 W (0.44 kW)**, assuming the leak runs continuously.

Period	Energy lost	Bill at 4.5 THB/kWh	Bill at 4.7 THB/kWh
1 hour	0.44 kWh	2.0 THB	2.1 THB
1 day (24 h)	10.5 kWh	47 THB	50 THB
1 month (30 days)	**316 kWh**	**1,420 THB**	**1,490 THB**
1 year	3,850 kWh	17,300 THB	18,100 THB



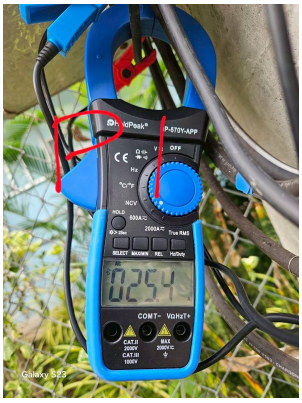
At 220 V the loss is about 420 W (≈ 1,360–1,420 THB a month). If the leak comes from one appliance that only runs part of the day, the loss is lower. During sunny hours the leak is fed by solar, so that share does not show on the PEA bill but is still lost solar energy. Rates include Ft and VAT; we will recalculate from the site's actual PEA bill.

Safety



Electric-shock risk
Either this supply has no residual-current protection (RCD/RCBO), or current is returning through earth or another circuit's neutral. Both can give an electric shock – important in a nursing home with elderly residents.

Reference: phase currents at commissioning (Sept 2025)



P1: 25.4 A



P2: 13.0 A



P3: 8.7 A

These are normal load currents per phase, recorded when the system was handed over. They are not leakage readings. After the leak is repaired, the same points will be measured again, together with a residual (L + N) reading that should be close to zero.

Next steps for the leakage

1. Switch circuits off one at a time while watching the clamp meter to find the leaking circuit or appliance (water heaters, pumps and old wiring are common causes).
2. Insulation-test that circuit and repair or replace the faulty part.
3. Make sure RCD protection is installed on the supply; add it where missing.
4. Re-measure after the repair to confirm the leakage is close to zero.

Action plan

The solar system itself is in good condition. The site's own power use and its electrical infrastructure need monitoring and some correction.

Priority	Action	By
HIGH	Trace and repair the ~2 A leakage; confirm RCD protection	Solar ESS with the site's electrician
HIGH	Re-terminate or replace the hot neutral cable at the old meter (65.4 °C today)	Solar ESS
MEDIUM	Decide: replace or remove the cracked module (Solar ESS to quote 1 module + installation)	Customer → Solar ESS
ROUTINE	Re-scan joints and re-measure leakage after the repairs	Solar ESS
ROUTINE	Review daytime loads on SolisCloud to use more of the solar output	Solar ESS with the customer

Contact
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All qualifications and certificates will be furnished upon request or upon completion of the contract.

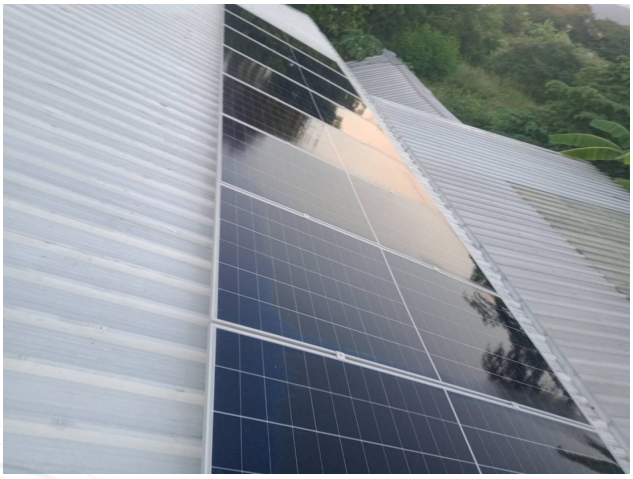
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After cleaning – 14 November 2026

All PV modules were cleaned during today's visit. The photos below show the array after cleaning.



Rows after cleaning



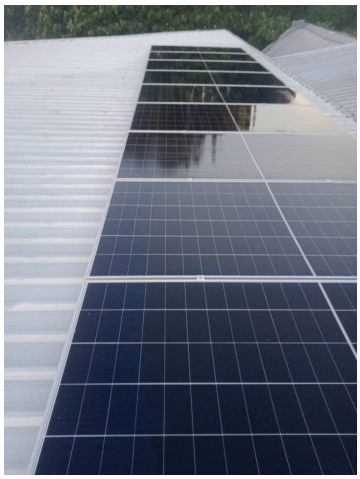
Clean glass along the ridge



Rear rows after cleaning



Array overview after cleaning



Module surface after cleaning



Upper rows after cleaning

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